

МУҲАММАД АЛ-ХОРАЗМИЙ НОМИДАГИ ТОШКЕНТ АХБОРОТ ТЕХНОЛОГИЯЛАРИ УНИВЕРСИТЕТИ

Мултимедиа технологиялари кафедраси

**CSS3 асослари: эффектлар, трансформация,
анимация. CSS3 ёрдамида HTML
саҳифаларни таҳрирлаш.**

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Режа:

1. CSS матн эффектлари
2. CSS да 2 ўлчовли трансформация
3. CSS да 3 ўлчовли трансформация
4. CSS анимациялар

CSS матн эффе́ктлари

Хусусият	Изоҳ
text-align-last	Матннинг охирги сатри қандай текисланиши кераклигини белгилайди
text-justify	Матннинг қандай текисланиши ва интервалда жойлаштирилишини белгилайди
text-overflow	Матн ҳажми сиғмасдан кўрсатилмай қолганда фойдаланувчини қандай хабардор қилишни белгилайди
word-break	Скрипт иштирок этмаган матнни қандай ажратилишини белгилайди
word-wrap	Уzun сўзларни кейинги сатрга кўчириш белгилайди
writing-mode	Матн сатрларини горизонтал ва вертикал ҳолатда жойлашувини белгилайди

CSS матем эффектлари: text-align-last

```
text-align-last: auto|left|right|center|justify|start|end|  
initial|inherit;
```

```
div.a {  
    text-align: justify; /* For Edge */  
    -moz-text-align-last: right; /* For  
Firefox prior 58.0 */  
    text-align-last: right;  
}  
  
div.b {  
    text-align: justify; /* For Edge */  
    -moz-text-align-last: center; /* For  
Firefox prior 58.0 */  
    text-align-last: center;  
}  
  
div.c {  
    text-align: justify; /* For Edge */  
    -moz-text-align-last: justify; /* For  
Firefox prior 58.0 */  
    text-align-last: justify;  
}
```

text-align-last: right:

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam semper diam at erat pulvinar, at pulvinar felis blandit. Vestibulum volutpat tellus diam, consequat gravida libero rhoncus ut.

text-align-last: center:

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam semper diam at erat pulvinar, at pulvinar felis blandit. Vestibulum volutpat tellus diam, consequat gravida libero rhoncus ut.

text-align-last: justify:

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam semper diam at erat pulvinar, at pulvinar felis blandit. Vestibulum volutpat tellus diam, consequat gravida libero rhoncus ut.

CSS математик эффектлари: text-justify

```
text-justify: auto|inter-word|inter-character|none|initial|  
inherit;
```

```
div {  
  text-align: justify;  
  text-justify: inter-word;  
}
```

The text-justify Property

In my younger and more vulnerable years my father gave me some advice that I've been turning over in my mind ever since. 'Whenever you feel like criticizing anyone,' he told me, 'just remember that all the people in this world haven't had the advantages that you've had.'

CSS мaтн эффeктлaри: text-overflow

```
text-overflow: clip|ellipsis|string|initial|inherit;
```

```
div.a {  
  white-space: nowrap;  
  width: 100px;  
  overflow: hidden;  
  text-overflow: ellipsis;  
  border: 1px solid #000000;  
}  
div.b {  
  white-space: nowrap;  
  width: 100px;  
  overflow: hidden;  
  text-overflow: clip;  
  border: 1px solid #000000;  
}
```

This is some...

This is some lo

This is some long text that will not fit in the box.

CSS матн эффектлари: word-break

```
word-break: normal|break-all|keep-all|break-word|initial|inherit;
```

```
<style>
p {
  width: 140px;
  border: 1px solid #000000;
}
p.a {
  word-break: normal;
}
p.b {
  word-break: keep-all;
}
p.c {
  word-break: break-all;
}
</style>
```

word-break: normal (default):

Thisissomeveryveryverylong
word. Words will
break according to
usual rules.

word-break: keep-all:

Thisissomeveryveryverylong
word. This text
will-break-at-hyphens.

word-break: break-all:

Thisissomeveryveryv
erylong word. This te
xt will break at any c
haracter.

CSS матн эффектлари: word-wrap

```
word-wrap: normal|break-word|initial|inherit;
```

```
div {  
  width: 150px;  
  border: 1px solid #000000;  
}  
  
div.a {  
  word-wrap: normal;  
}  
  
div.b {  
  word-wrap: break-word;  
}
```

word-wrap: normal (default):

This div contains a very long word:
thisisaveryveryveryveryveryverylongword.
The long word will break and wrap to the next line.

word-wrap: break-word:

This div contains a very long word:
thisisaveryveryveryveryveryverylongword.
The long word will break and wrap to the next line.

CSS мaтн эффeктлaри: writing-mode

```
writing-mode: horizontal-tb|vertical-rl|vertical-lr;
```

```
p.test1 {  
  writing-mode: horizontal-tb;  
}
```

```
p.test2 {  
  writing-mode: vertical-rl;  
}
```

```
span.test2 {  
  writing-mode: vertical-rl;  
}
```

The writing-mode Property

Some text with default writing-mode.

Some text with a span element with a vertical-rl writing-mode.

Some writing-mode: vertical-rl.

CSS да 2 ўлчовли трансформация

Хусусият	Изоҳ
transform	Элементга икки ва уч ўлчовли трансформацияни қўллайди.
transform-origin	Трансформация қилинаётган элементлар ҳолатини ўзгартириш имкониятини беради

Функция	Изоҳ
matrix(n,n,n,n,n,n)	Олти қийматли матрицадан фойдаланиб, 2 ўлчовли трансформация ҳосил қилинади.
translate(x,y)	Элементни X ва Y ўқи бўйлаб 2 ўлчовли кўчиришни амалга оширади.
translateX(n)	Элементни X ўқи бўйлаб 2 ўлчовли кўчиришни амалга оширади.
translateY(n)	Элементни Y ўқи бўйлаб 2 ўлчовли кўчиришни амалга оширади.
scale(x,y)	2 ўлчовли трансформациянинг кенглиги ва баландлигини ўзгартиришни амалга оширади.
scaleX(n)	2 ўлчовли трансформациянинг кенглиги ўзгартиришни амалга оширади.
scaleY(n)	2 ўлчовли трансформациянинг баландлигини ўзгартиришни амалга оширади.
rotate(angle)	Маълум бурчак асосида 2 ўлчовли айлантиришни амалга оширади.
skew(x-angle,y-angle)	Элементни X ва Y ўқи бўйлаб 2 ўлчовли айлантиришни амалга оширади.
skewX(angle)	Элементни X ўқи бўйлаб 2 ўлчовли айлантиришни амалга оширади.
skewY(angle)	Элементни Y ўқи бўйлаб 2 ўлчовли айлантиришни амалга оширади.

CSS да 2 үлчовли трансформация: transform

transform: none|transform-functions|initial|inherit;

```
div.a {
  width: 100px;
  height: 80px;
  background-color: yellow;
  -ms-transform: rotate(20deg);
/* IE 9 */
  -webkit-transform: rotate(20deg);
/* Safari 3-8 */
  transform: rotate(20deg);
}
div.b {
  width: 100px;
  height: 80px;
  background-color: yellow;
  -ms-transform: skewY(20deg);
/* IE 9 */
  -webkit-transform: skewY(20deg);
/* Safari 3-8 */
  transform: skewY(20deg);
}
```

transform: rotate(20deg):



transform: skewY(20deg):



CSS да 2 үлчовли трансформация: transform

```
div.c {  
  width: 100px;  
  height: 80px;  
  background-color: yellow;  
  -ms-transform: scaleY(1.5); /* IE 9 */  
  -webkit-transform: scaleY(1.5); /*  
Safari 3-8 */  
  transform: scaleY(1.5);  
}
```

transform: scaleY(1.5):

Hello World!

CSS да 2 ўлчовли трансформация: transform-origin

```
transform-origin: x-axis y-axis z-axis|initial|inherit;
```

Қиймат	Изоҳ
x-axis	X ўқи бўйлаб жойлашувини аниқлайди. left, center, right, <i>length</i> , % ларни қабул қилади.
y-axis	Y ўқи бўйлаб жойлашувини аниқлайди. top, center, bottom, <i>length</i> , % ларни қабул қилади.
z-axis	3 ўлчовли трансформацияда Z ўқи бўйлаб жойлашувини аниқлайди. <i>length</i> ни қабул қилади.

CSS да 2 үлчовли трансформация: matrix(n,n,n,n,n,n)

`matrix(scaleX() ,skewY() ,skewX() ,scaleY() ,translateX() ,
translateY())`

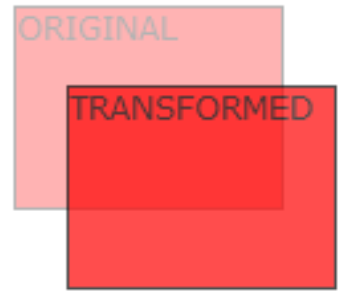
```
div {  
  width: 300px;  
  height: 100px;  
  background-color: yellow;  
  border: 1px solid black;  
}  
div#myDiv1 {  
  -ms-transform: matrix(1, -0.3, 0, 1, 0, 0); /*  
  IE 9 */  
  -webkit-transform: matrix(1, -0.3, 0, 1, 0, 0);  
  /* Safari prior 9.0 */  
  transform: matrix(1, -0.3, 0, 1, 0, 0);  
  /* Standard syntax */  
}  
div#myDiv2 {  
  -ms-transform: matrix(1, 0, 0.5, 1, 150, 0); /*  
  IE 9 */  
  -webkit-transform: matrix(1, 0, 0.5, 1, 150, 0);  
  /* Safari prior 9.0 */  
  transform: matrix(1, 0, 0.5, 1, 150, 0);  
  /* Standard syntax */  
}
```

This a normal div element.

Using the matrix() method.

Another use of the matrix() method.

CSS да 2 ўлчовли трансформация: translate(x,y)



```
div {  
  width: 300px;  
  height: 100px;  
  background-color: yellow;  
  border: 1px solid black;  
  -ms-transform: translate(50px,100px); /* IE 9 */  
  -webkit-transform: translate(50px,100px); /* Safari prior 9.0 */  
  transform: translate(50px,100px); /* Standard syntax */  
}
```

This div element is moved 50 pixels to the right, and 100 pixels down from its current position.

CSS да 2 үлчовли трансформация: scale(x,y)



```
div {  
  margin: 150px;  
  width: 200px;  
  height: 100px;  
  background-color: yellow;  
  border: 1px solid black;  
  -ms-transform: scale(2,3); /* IE 9 */  
  -webkit-transform: scale(2,3); /* Safari prior 9.0 */  
  transform: scale(2,3); /* Standard syntax */  
}
```

This div element is two times of its original width, and three times of its original height.

CSS да 2 ўлчовли трансформация: rotate(angle)



```
div {  
  width: 300px;  
  height: 100px;  
  background-color: yellow;  
  border: 1px solid black;  
}  
  
div#myDiv {  
  -ms-transform: rotate(20deg); /* IE 9 */  
  -webkit-transform: rotate(20deg); /* Safari prior 9.0 */  
  transform: rotate(20deg); /* Standard syntax */  
}
```

This a normal div element.

This div element is rotated clockwise 20 degrees.

CSS да 2 ўлчовли трансформация: skewX(angle), skewY(angle)

```
div {  
  width: 300px;  
  height: 100px;  
  background-color: yellow;  
  border: 1px solid black;  
}  
div#myDiv {  
  -ms-transform: skewX(20deg); /* IE 9 */  
  -webkit-transform: skewX(20deg); /* Safari prior 9.0 */  
  transform: skewX(20deg); /* Standard syntax */  
}  
div#myDiv2 {  
  -ms-transform: skewY(20deg); /* IE 9 */  
  -webkit-transform: skewY(20deg); /* Safari prior 9.0 */  
  transform: skewY(20deg); /* Standard syntax */  
}
```

This a normal div element.

This div element is skewed 20 degrees along the X-axis.

This div element is skewed 20 degrees along the X-axis.

CSS да 3 ўлчовли трансформация

Хусусият	Изоҳ
transform	Элементга икки ва уч ўлчовли трансформацияни қўллайди.
transform-origin	Трансформация қилинаётган элементлар ҳолатини ўзгартириш имкониятини беради
transform-style	Ичма-ич элементларнинг 3 ўлчовли фазода қандай кўрсатилишини белгилайди
perspective	3 ўлчовли элементларнинг аксонометрик кўринишини белгилайди.
perspective-origin	3 ўлчовли элементларнинг пастки жойлашувини (позициясини) белгилайди.
backface-visibility	Экранга қарама-қарши бўлганда элемент кўриниши ёки кўринмаслигини белгилайди

CSS да 3 ўлчовли трансформация

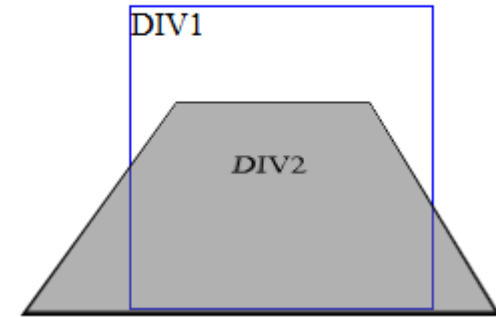
Функция	Изоҳ
matrix3d (<i>n,n,n,n,n,n,n,n,n,n,n,n,n,n,n,n</i>)	16 қийматли 4x4 матрицадан фойдаланиб, 3 ўлчовли трансформация ҳосил қилинади.
translate3d(x,y,z)	Элементни 3 ўлчовли кўчиришни амалга оширади.
translateX(x)	Элементни X ўқи бўйлаб 3 ўлчовли кўчиришни амалга оширади.
translateY(y)	Элементни Y ўқи бўйлаб 3 ўлчовли кўчиришни амалга оширади.
translateZ(z)	Элементни Z ўқи бўйлаб 3 ўлчовли кўчиришни амалга оширади
scale3d(x,y,z)	3 ўлчовли трансформациянинг масштабини ўзгартиришни амалга оширади.
scaleX(x)	3 ўлчовли трансформациянинг масштабини X ўқи бўйлаб ўзгартиришни амалга оширади.
scaleY(y)	3 ўлчовли трансформациянинг масштабини Y ўқи бўйлаб ўзгартиришни амалга оширади.
scaleZ(z)	3 ўлчовли трансформациянинг масштабини Z ўқи бўйлаб ўзгартиришни амалга оширади.
rotate3d(x,y,z,angle)	Маълум бурчак асосида 3 ўлчовли айлантиришни амалга оширади.
rotateX(angle)	Элементни X ўқи бўйлаб 3 ўлчовли айлантиришни амалга оширади.
rotateY(angle)	Элементни Y ўқи бўйлаб 3 ўлчовли айлантиришни амалга оширади.
rotateZ(angle)	Элементни Z ўқи бўйлаб 3 ўлчовли айлантиришни амалга оширади.

CSS да 3 үлчовли трансформация: perspective

perspective: length|none;

```
<style>
#div1 {
  position: relative;
  height: 150px; width: 150px;
  margin-left: 60px; border: 1px solid blue;
  -webkit-perspective: 100px; /* Safari 4-8 */
  perspective: 100px;
}
#div2{
  padding: 50px; position: absolute;
  border: 1px solid black; background-color:
red;
  background: rgba(100,100,100,0.5);
  -webkit-transform-style: preserve-3d; /* Safari
3-8 */
  -webkit-transform: rotateX(45deg); /* Safari 3-
8 */
  transform-style: preserve-3d;
  transform: rotateX(45deg);
}
</style>
<h2>perspective: 100px:</h2>
<div id="div1">DIV1
  <div id="div2">DIV2</div>
</div>
```

perspective: 100px:

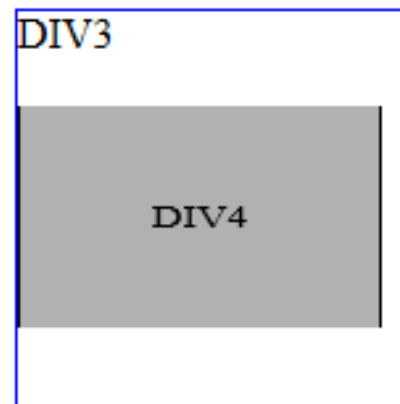


CSS да 3 үлчовли трансформация: perspective

```
<style>
#div4 {
  padding: 50px; position: absolute;
  border: 1px solid black;
  background-color: red;
  background: rgba(100,100,100,0.5);
  -webkit-transform-style: preserve-3d; /*
Safari 3-8 */
  -webkit-transform: rotateX(45deg); /* Safari
3-8 */
  transform-style: preserve-3d;
  transform: rotateX(45deg);
}
#div3 {
  position: relative; height: 150px;
  width: 150px; margin-left: 60px;
  border: 1px solid blue;
  -webkit-perspective: none; /* Safari 4-8 */
  perspective: none;
}</style>
```

```
<h2>perspective: none:</h2>
<div id="div3">DIV3
  <div id="div4">DIV4</div>
</div>
```

perspective: none:



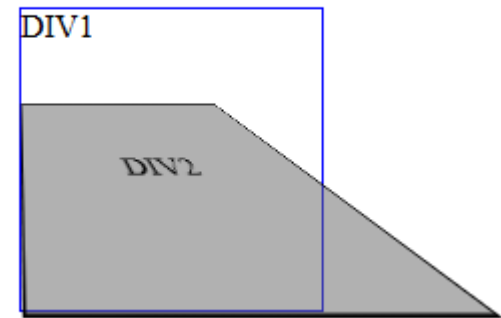
CSS да 3 үлчовли трансформация: perspective-origin

perspective-origin: x-axis y-axis|initial|inherit;

```
<style>
#div1 {
  position: relative; height: 150px; width: 150px;
  margin-left: 60px; border: 1px solid blue;
  -webkit-perspective: 100px; /* Safari 4-8 */
  -webkit-perspective-origin: left; /* Safari 4-8 */
  perspective: 100px;
  perspective-origin: left;
}
#div2{
  padding: 50px; position: absolute;
  border: 1px solid black; background-color: red;
  background: rgba(100,100,100,0.5);
  -webkit-transform-style: preserve-3d; /* Safari 3-8 */
  -webkit-transform: rotateX(45deg); /* Safari 3-8 */
  transform-style: preserve-3d;
  transform: rotateX(45deg);
}

</style>
<h2>perspective-origin: left:</h2>
<div id="div1">DIV1
  <div id="div2">DIV2</div>
</div>
```

perspective-origin: left:



CSS да 3 үлчовли трансформация: backface-visibility

backface-visibility: visible|hidden|initial|inherit;

```
div {
  position: relative;
  height: 60px;
  width: 60px;
  background-color: red;
  -webkit-transform: rotateY(180deg); /* Chrome, Safari,
Opera */
  transform: rotateY(180deg);
}
#div1 {
  -webkit-backface-visibility: hidden; /* Chrome, Safari,
Opera */
  backface-visibility: hidden;
}
#div2 {
  -webkit-backface-visibility: visible; /* Chrome,
Safari, Opera */
  backface-visibility: visible;
}
```

This div element has "backface-visibility: hidden", and the back face of the div element is invisible:

This div element has "backface-visibility: visible", and the back face of the div element shows a mirror image of the front face:



CSS анимациялар

CSS да JavaScript ва Flash лардан фойдаланмаган ҳолда HTML элементларга анимацияларни қўллаш мумкин:

Хусусият	Изоҳ
@keyframes	Анимация кодини белгилайди
animation	Барча анимация хусусиятларини бир вақтда ўрнатишни белгилайди
animation-delay	Анимация бошланишининг кечикишини белгилайди
animation-direction	Анимациянинг олдинга, орқага ва навбвт билан амалга оширилишини белгилайди
animation-duration	Анимациянинг бир марта бажарилиш цикли учун қанча вақт кетишини белгилайди
animation-fill-mode	Анимациянинг бажарилмай турган ҳолатда (бошланишидан олдин, бажарилиб бўлгандан сўнг ёки ҳар икки ҳолатда) элементнинг стилини белгилайди
animation-iteration-count	Анимациянинг неча марта бажарилиш кераклигини белгилайди
animation-name	@keyframes анимациянинг номини белгилайди
animation-play-state	Анимациянинг бажарилаётганлиги ёки пауза қилинганлигини белгилайди
animation-timing-function	Анимациянинг эгри чизиқли тезлигини белгилайди

CSS анимациялар: animation

```
animation: name duration timing-function delay iteration-  
count direction fill-mode play-state;
```

```
div {  
  width: 100px;  
  height: 100px;  
  background: red;  
  position: relative;  
  -webkit-animation: mymove 5s infinite; /*  
Safari 4.0 - 8.0 */  
  animation: mymove 5s infinite;  
}  
  
/* Safari 4.0 - 8.0 */  
@-webkit-keyframes mymove {  
  from {left: 0px;}  
  to {left: 200px;}  
}  
  
@keyframes mymove {  
  from {left: 0px;}  
  to {left: 200px;}  
}
```

Натижа

```
div {  
  animation-name: example;  
  animation-duration: 5s;  
  animation-timing-function:  
linear;  
  animation-delay: 2s;  
  animation-iteration-count:  
infinite;  
  animation-direction:  
alternate;  
}
```



```
div {  
  animation: example 5s linear  
2s infinite alternate;  
}
```

CSS анимациялар: @keyframes

```
@keyframes animationname {keyframes-selector {css-styles;}}
```

Қиймат	Изоҳ
animationname	Мажбурий қиймат ҳисобланади. Анимация номини белгилайди
keyframes-selector	Мажбурий қиймат ҳисобланади. Анимация давомийлигининг % кўрсаткичи ҳисобланади (0-100%); from (0%), to (100%)
css-styles	Мажбурий қиймат ҳисобланади. Бир ёки бир нечта CSS хусусиятларни белгилаш мумкин

CSS анимациялар: @keyframes

```
div {
  width: 100px;
  height: 100px;
  background: red;
  position :relative;
  -webkit-animation: mymove 5s infinite; /* Safari 4.0
- 8.0 */
  animation: mymove 5s infinite;
}
/* Safari 4.0 - 8.0 */
@-webkit-keyframes mymove {
  from {top: 0px;}
  50% {top: 100px !important}
  to {top: 200px;}
}
/* Standard syntax */
@keyframes mymove {
  from {top: 0px;}
  50% {top: 100px !important} /* ignored */
  to {top: 200px;}
}
```

[Натижа](#)

CSS анимациялар: @keyframes

```
<style>
div {
  width: 100px; height: 100px;
  background: red; position: relative;
  -webkit-animation: mymove 5s infinite; /*
Safari 4.0 - 8.0 */
  animation: mymove 5s infinite;
}
/* Safari 4.0 - 8.0 */
@-webkit-keyframes mymove {
  0%    {top: 0px;}
  25%   {top: 200px;}
  75%   {top: 50px}
  100%  {top: 100px;}
}
/* Standard syntax */
@keyframes mymove {
  0%    {top: 0px;}
  25%   {top: 200px;}
  75%   {top: 50px}
  100%  {top: 100px;}
}
```

[Натижа](#)

CSS анимациялар: @keyframes

```
div {  
  width: 100px; height: 100px;  
  background: red; position: relative;  
  -webkit-animation: mymove 5s infinite; /* Safari 4.0 - 8.0 */  
  animation: mymove 5s infinite;  
}  
/* Safari 4.0 - 8.0 */  
@-webkit-keyframes mymove {  
  0%    {top: 0px; background: red; width: 100px;}  
  100%  {top: 200px; background: yellow; width: 300px;}  
}  
/* Standard syntax */  
@keyframes mymove {  
  0%    {top: 0px; background: red; width: 100px;}  
  100%  {top: 200px; background: yellow; width: 300px;}  
}
```

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CSS анимациялар: @keyframes

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```
div {
  width: 100px; height: 100px;
  background: red; position: relative;
  -webkit-animation: mymove 5s infinite; /* Safari 4.0 - 8.0 */
  animation: mymove 5s infinite;
}
/* Safari 4.0 - 8.0 */
@-webkit-keyframes mymove {
  0%    {top: 0px; left: 0px; background: red;}
  25%   {top: 0px; left: 100px; background: blue;}
  50%   {top: 100px; left: 100px; background: yellow;}
  75%   {top: 100px; left: 0px; background: green;}
  100%  {top: 0px; left: 0px; background: red;}
}
/* Standard syntax */
@keyframes mymove {
  0%    {top: 0px; left: 0px; background: red;}
  25%   {top: 0px; left: 100px; background: blue;}
  50%   {top: 100px; left: 100px; background: yellow;}
  75%   {top: 100px; left: 0px; background: green;}
  100%  {top: 0px; left: 0px; background: red;}
}
```


CSS анимациялар: animation-delay

```
animation-delay: time|initial|inherit;
```

```
div {  
  width: 100px; height: 100px;  
  background: red; position: relative;  
  -webkit-animation: mymove 5s infinite; /*  
Safari 4.0 - 8.0 */  
  -webkit-animation-delay: 2s; /* Safari 4.0 -  
8.0 */  
  animation: mymove 5s infinite;  
  animation-delay: 2s;  
}  
/* Safari 4.0 - 8.0 */  
@-webkit-keyframes mymove {  
  from {left: 0px;}  
  to {left: 200px;}  
}  
@keyframes mymove {  
  from {left: 0px;}  
  to {left: 200px;}  
}
```

[Натижа](#)

CSS анимациялар: animation-direction

```
animation-direction: normal|reverse|alternate|alternate-reverse|initial|inherit;
```

Қиймат	Изоҳ
normal	Одатдаги (сукут ҳолатдаги) қиймат. Анимация нормал шаклда ишга туширилади (олдинга қараб)
reverse	Анимация тескари йўналишда ишга туширилади (орқага қараб)
alternate	Анимация аввал олдинга, кейин орқага йўналтирилган ҳолда ишга туширилади
alternate-reverse	Анимация аввал орқага, кейин олдинга йўналтирилган ҳолда ишга туширилади

CSS анимациялар: animation-direction

[Натижа](#)

```
div {
  width: 100px; height: 100px;
  background-color: red; position: relative;
  -webkit-animation-name: example; /* Safari 4.0 - 8.0 */
  -webkit-animation-duration: 4s; /* Safari 4.0 - 8.0 */
  -webkit-animation-iteration-count: 2; /* Safari 4.0 - 8.0 */
  -webkit-animation-direction: alternate; /* Safari 4.0 - 8.0 */
  animation-name: example; animation-duration: 4s;
  animation-iteration-count: 2; animation-direction: alternate;
}

/* Safari 4.0 - 8.0 */
@-webkit-keyframes example {
  0%    {background-color:red; left:0px; top:0px;}
  25%   {background-color:yellow; left:200px; top:0px;}
  50%   {background-color:blue; left:200px; top:200px;}
  75%   {background-color:green; left:0px; top:200px;}
  100%  {background-color:red; left:0px; top:0px;}
}

/* Standard syntax */
@keyframes example {
  0%    {background-color:red; left:0px; top:0px;}
  25%   {background-color:yellow; left:200px; top:0px;}
  50%   {background-color:blue; left:200px; top:200px;}
  75%   {background-color:green; left:0px; top:200px;}
  100%  {background-color:red; left:0px; top:0px;}
}
```

CSS анимациялар: animation-duration

```
animation-duration: time|initial|inherit;
```

```
div {  
  width: 100px; height: 100px;  
  background: red; position: relative;  
  -webkit-animation: mymove infinite; /* Safari  
4.0 - 8.0 */  
  -webkit-animation-duration: 3s; /* Safari 4.0  
- 8.0 */  
  animation: mymove infinite;  
  animation-duration: 3s;  
}  
/* Safari 4.0 - 8.0 */  
@-webkit-keyframes mymove {  
  from {top: 0px;}  
  to {top: 200px;}  
}  
@keyframes mymove {  
  from {top: 0px;}  
  to {top: 200px;}  
}
```

[Натижа](#)

CSS анимациялар: animation-iteration-count

`animation-iteration-count: number|infinite|initial|inherit;`

```
div {
  width: 100px; height: 100px;
  background: red; position: relative;
  -webkit-animation: mymove 3s; /* Safari 4.0
- 8.0 */
  -webkit-animation-iteration-count: 2; /*
Safari 4.0 - 8.0 */
  animation: mymove 3s;
  animation-iteration-count: 2;
}
/* Safari 4.0 - 8.0 */
@-webkit-keyframes mymove {
  from {top: 0px;}
  to {top: 200px;}
}
@keyframes mymove {
  from {top: 0px;}
  to {top: 200px;}
}
```

[Натижа](#)

CSS анимациялар: animation-name

`animation-name: keyframename|none|initial|inherit;`

```
div {
  width: 100px;
  height: 100px;
  background: red;
  position: relative;
  -webkit-animation-name: mymove; /* Safari 4.0
- 8.0 */
  -webkit-animation-duration: 5s; /* Safari 4.0
- 8.0 */
  animation-name: mymove;
  animation-duration: 5s;
}
/* Safari 4.0 - 8.0 */
@-webkit-keyframes mymove {
  from {left: 0px;}
  to {left: 200px;}
}
@keyframes mymove {
  from {left: 0px;}
  to {left: 200px;}
}
```

[Натижа](#)

CSS анимациялар: animation-play-state

`animation-play-state: paused|running|initial|inherit;`

```
div {
  width: 100px;
  height: 100px;
  background: red;
  position: relative;
  -webkit-animation: mymove 5s; /* Safari 4.0 -
8.0 */
  -webkit-animation-play-state: paused; /*
Safari 4.0 - 8.0 */
  animation: mymove 5s;
  animation-play-state: paused;
}
/* Safari 4.0 - 8.0 */
@-webkit-keyframes mymove {
  from {left: 0px;}
  to {left: 200px;}
}
@keyframes mymove {
  from {left: 0px;}
  to {left: 200px;}
}
```

[Натижа](#)

CSS анимациялар: animation-play-state

```
div {
  width: 100px;
  height: 100px;
  background: red;
  position: relative;
  -webkit-animation: mymove 5s infinite; /*
Safari 4.0 - 8.0 */
  animation: mymove 5s infinite;
}
div:hover {
  -webkit-animation-play-state: paused; /*
Safari 4.0 - 8.0 */
  animation-play-state: paused;
}
/* Safari 4.0 - 8.0 */
@-webkit-keyframes mymove {
  from {left: 0px;}
  to {left: 200px;}
}
@keyframes mymove {
  from {left: 0px;}
  to {left: 200px;}
}
```

[Натижа](#)

CSS анимациялар: animation-timing-function

```
animation-timing-function: linear|ease|ease-in|ease-out|ease-in-out|step-start|step-end|steps(int,start|end)|cubic-bezier(n,n,n,n)|initial|inherit;
```

Қиймат	Изоҳ
linear	Анимация бошидан охиригача бир хил тезликка эга бўлишини белгилайди
ease	Сукут ҳолатдаги қиймат. Анимация секин бошланади, кейин тезлашади, тугашидан олдин секинлашади.
ease-in	Анимация секин бошланади
ease-out	Анимациянинг охири секин тугайди
ease-in-out	Анимация секин бошланиб, секин тугайди
step-start	step функциясига ўхшаш.
step-end	step функциясига ўхшаш.
steps(int,start end)	Иккита параметрли қадамни белгилаш функцияси. 1-параметр интерваллар сонини белгилайди (0 дан катта бутун сон қабул қилади).
cubic-bezier(n,n,n,n)	Кубик Безье функцияси қийматларни белгилайди. 0 дан 1 гача қиймат қабул қилади

CSS анимациялар: animation-play-state

```
#div1 {animation-timing-function: linear;}  
#div2 {animation-timing-function: ease;}  
#div3 {animation-timing-function: ease-in;}  
#div4 {animation-timing-function: ease-out;}  
#div5 {animation-timing-function: ease-in-out;}
```

[Натижа](#)

```
#div1 {animation-timing-function: cubic-bezier(0,0,1,1);}  
#div2 {animation-timing-function: cubic-bezier(0.25,0.1,0.25,1);}  
#div3 {animation-timing-function: cubic-bezier(0.42,0,1,1);}  
#div4 {animation-timing-function: cubic-bezier(0,0,0.58,1);}  
#div5 {animation-timing-function: cubic-bezier(0.42,0,0.58,1);}
```

[Натижа](#)